AudioLLM-D: Fine-tuning Audio LLM for Dementia speech

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Abstract

Early diagnosis of dementia is a significant challenge. General-purpose Voice AI models often suppress language disfluencies such as pauses and filler words, which serve as critical biomarkers for cognitive decline, resulting in both the digital exclusion of People with Dementia (PwD) and missed diagnostic opportunities. Existing efforts remain fragmented, with no integrated Audio LLM that simultaneously understands PwD speech, reasons, and facilitates diagnostic interactions. We bridge this gap by developing a dementia-specific Multimodal LLM, fine-tuning Qwen2-Audio-7B-Instruct on the DementiaBank Pitt Corpus. Our primary goal is accurate interpretation of the voice and intent of PwDs to ensure accessibility. As a stretch goal, we will extend the system toward diagnosis through adaptive, clinically-informed conversations.

1 Introduction

Dementia affects over 57 million people worldwide, yet an estimated 75% of cases go undiagnosed. Speech offers a compelling non-invasive biomarker: persons with dementia (PwDs) exhibit characteristic disfluencies including filled pauses (uh, um), word-finding hesitations, repetitions, and fragmented sentences that worsen with cognitive decline. General-purpose Voice AI models are trained to filter out these disfluencies as noise, discarding the very signals critical for diagnosis. This creates a two-fold problem. First, voice-based models fail to interpret the intent of PwDs, cutting them off from modern AI tools. Second, models smooth over the speech patterns that constitute essential evidence for reasoning about diagnosis. WhisperD [1] showed that fine-tuning Whisper [4] improves dementia transcription, but its encoder-decoder architecture cannot reason or classify. We propose an integrated dementia-specific Multimodal LLM, fine-tuned on dementia speech corpus. The model would understand PwD speech, reason, and facilitate diagnostic interactions, enabling scalable early screening.

2 Approach

We train and evaluate on the DementiaBank Pitt Corpus [2], which contains audio with CHAT-format transcripts from 104 healthy controls and 208 participants with dementia, using a speaker-disjoint 80/10/10 split and 30-second segments aligned via Montreal Forced Aligner. Our model builds on Qwen2-Audio-7B-Instruct [3], a Whisper-large-v3 encoder [4] paired with a Qwen-7B language model (LM). We extend its tokenizer with three disfluency tokens (<short_pause>, <medium_pause>, <long_pause>) for gaps of 0.3–1.0s, 1.0–2.5s, and >2.5s, initialized from the mean of existing embeddings and kept trainable. Fine-tuning applies different learning rates across four components: LoRA on the LM, LoRA on the upper audio encoder layers, full fine-tuning of the audio-to-language projection, and trainable input embedding and LM head layers for the new disfluency tokens. Disfluency tokens receive upweighted loss to compensate for their low frequency. We evaluate along two axes. First, transcription quality on raw audio against reference transcripts, benchmarked against WhisperD. Second, diagnostic reasoning capability assessed via QA pairs generated by an LLM from the reference transcripts. As a stretch goal, we will extend the QA evaluation toward multi-turn diagnostic interaction.

3 Project Timeline

Week 1: Prepare the dataset and setup baselines.
Week 2: Fine-tune Qwen2-Audio-7B-Instruct with audio and transcribed text.
Week 3: Prepare QA eval dataset and Evaluate model performance.
Week 4: Iterate, refine, and write the final report.

References

[1] Emmanuel Akinrintoyo, Nadine Abdelhalim, and Nicole Salomons. 2025. WhisperD: Dementia speech recognition and filler word detection with Whisper. In *Interspeech*.

[2] James T Becker, François Boller, Oscar L Lopez, Judith Saxton, and Kelly L McGonigle. 1994. The natural history of Alzheimer's disease: Description of study cohort and accuracy of diagnosis. *Archives of Neurology*, 51(6):585–594.

[3] Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, Yichong Leng, Yuanjun Lv, Jinzheng He, Junyang Lin, Chang Zhou, and Jingren Zhou. 2024. Qwen2-audio technical report.

[4] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2022. Robust speech recognition via large-scale weak supervision.